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DAY 6

Turning SysML models into engineering reports

From SysML v2 to a readable design report

Concept design / PoC realizations / Traceable evidence

TODAY'S REPORT WORKFLOW

01 **CONFIRM**
Choose the model and sources

02 **EXTRACT**
Read docs and existing views

03 **REPORT**
Explain evidence and gaps

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DAY 6 / WHY GENERATE A REPORT

3 MIN

The model becomes the source of the report

Readable prose and diagrams should come from documented SysML, not from plausible AI guesses.

INPUT

SysML v2 model

Requirements, structure, behavior and documented design choices.

Concept and PoC models stay separate.

→

TRANSFORM

Model-report skill

Reads doc declarations; summarizes them as continuous prose; finds existing views.

→

OUTPUT

Markdown report

Keio-branded cover, two design parts, selected diagrams and source traceability.

The chat launches the skill; the SysML files provide the evidence.

DAY 6 / REPORT STRUCTURE

4 MIN

One report, two parts: WHAT and HOW

The report follows the same separation we use in the design methodology.

COVER + PROJECT SUMMARY

Keio logo • student • title • subtitle • summary

PART I / CONCEPT DESIGN

WHAT the product must do

1. **Requirements model**
2. **Structural model**
3. **Behavioral model**

PART II / PoC REALIZATIONS

HOW each alternative might realize it

- **PoC realization A / documented aspects**
- **PoC realization B / documented aspects**
- **Separate candidates; traceable to concept**

FINAL SECTIONS / Documentation and evidence gaps · Source traceability

DAY 6 / BEFORE THE REPORT

3 MIN

Confirm the model and discover its source files

The agent must establish scope before it reads, classifies or writes the report.

STUDENT CONFIRMS

Project + cover metadata

Confirm the concept model and PoC candidates.

Provide full name; approve the proposed report title and subtitle.

Agree on the source-file set.

ILLUSTRATIVE PROJECT FILE SET

models/project.sysml
models/project-realization.sysml
models/project-realization-alt.sysml

reports/project.md

Imported in-scope model files may be included. Ambiguous candidates require student confirmation.

DAY 6 / WHAT THE SKILL READS

4 MIN

The doc field gives the report its meaning

The agent explains documented semantics and names missing information rather than inventing it.

REQUIREMENTS

What the system must achieve

STRUCTURE

Logical parts and relationships

BEHAVIOR

Activities, interactions and flows

DOCUMENTED ELEMENT

Read the SysML doc text.
Rewrite as concise, continuous prose.
Cite the qualified element name.

MISSING OR UNCLEAR

No doc → mark undocumented.
No aspect → state not found.
Unverified claim → label as unverified.

Every paragraph must remain traceable to the source model.

DAY 6 / HOW THE DIAGRAMS APPEAR

4 MIN

Only existing views can become report diagrams

Each diagram is grouped by its actual content and inserted after the relevant section.

VIEW SELECTION + RENDERING

Existing SysML view → **Classify by exposed elements** → **Render using OpenSysML**

Only declared views are eligible. The agent never creates new views.

IF CONTENT IS MISSING

No view → no diagram

Render fails → record real error

Ambiguous view → ask student

Missing PoC → partial report

The report stays useful even when the model is incomplete.

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DAY 6 / END

Today, we made the model explain itself.

A report can communicate the design without inventing it.

Document the model. Prepare the views. Verify the evidence.

Questions?

TODAY'S REPORT MAKES VISIBLE

01 **CONCEPT**
What the product should do

02 **REALIZATIONS**
How a PoC may implement it

03 **GAPS**
What remains undocumented